ClinHalluEval: Measuring Hallucination in Clinical Language Models

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Abstract

ClinHalluEval extends FactProbe to clinical Medical QA. The evaluation suite contains 3,000 clinical questions derived from MedQA, with physician annotations for fabricated drugs, dosages, and interactions. It adopts the FactProbe metric while adding error categories that reflect clinical risk. We evaluate CuraLM-70B and two generic models. Even CuraLM-70B fabricates drug-interaction facts in about 14% of answers, and dosage errors are the most dangerous category. ClinHalluEval exposes failures that ordinary accuracy can hide, providing a safer basis for comparing a Large Language Model in clinical settings.

Keywords

Clinical hallucination, ClinHalluEval, MedQA, Hallucination Detection

1. Introduction

Accuracy alone cannot show whether a clinical answer contains a dangerous unsupported dosage or interaction. FactProbe [1] provides a useful annotation scheme and metric, while CuraLM-70B [2] is a strong adapted system whose behavior should be tested directly. Earlier work on structured drug-interaction facts [3] suggested a path to reducing fabricated statements, but evaluation has largely ignored it. ClinHalluEval fills that measurement gap.

2. Method

ClinHalluEval samples 3,000 questions from MedQA and expands each with a physician-reviewed answer key. Reviewers mark fabricated drugs, dosage claims, interactions, and unsupported certainty. The score is the proportion of answers with at least one fabricated clinical claim, reported overall and by category. We evaluate CuraLM-70B and two generic models with identical prompts and deterministic decoding. An adjudicator resolves disagreements between two physicians, and a separate audit estimates annotation agreement. The suite preserves FactProbe's span-level representation while adding a severity label for potential patient harm. The result is a clinical Hallucination Detection protocol that can be reused across systems without changing their generation code.

3. Experiments

CuraLM-70B has a 14.1% hallucination rate on ClinHalluEval, compared with 21.8% for the medium generic model and 27.5% for the smaller generic model. Dosage errors account for 38% of CuraLM-70B failures; drug-interaction fabrications account for 31%. Accuracy on the underlying MedQA subset remains high, confirming that answer correctness and safety are distinct.

System	Hallucination rate	Dosage-error share
Generic small	27.5%	42%
Generic medium	21.8%	40%
CuraLM-70B	14.1%	38%

4. Conclusion

ClinHalluEval makes clinical Hallucination Detection measurable at claim level. CuraLM-70B is strongest but still fabricates clinically important facts. The results renew interest in structured drug-interaction grounding and show that Medical QA evaluation must report both correctness and risk.

References

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